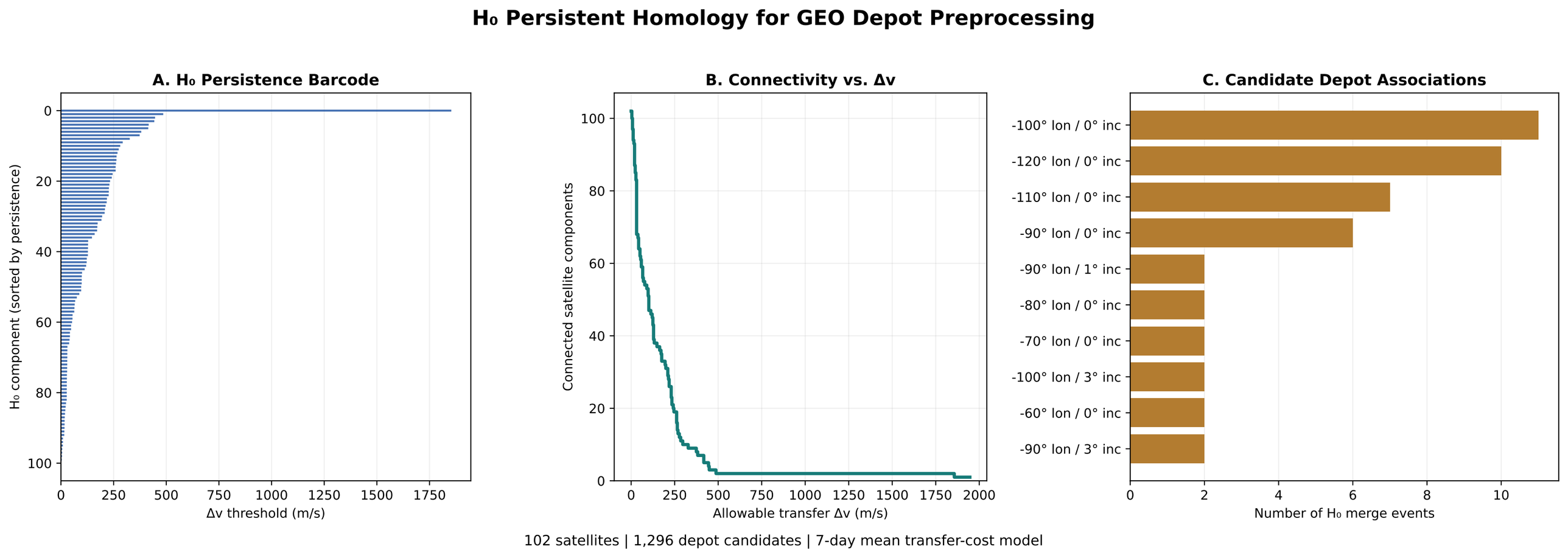
H₀ Persistent Homology for GEO Depot Preprocessing
A. H₀ Persistence Barcode
H₀ component (sorted by persistence)
Δv threshold (m/s)
B. Connectivity vs. Δv
Connected satellite components
Allowable transfer Δv (m/s)
C. Candidate Depot Associations
-100° lon / 0° inc
-120° lon / 0° inc
-110° lon / 0° inc
-90° lon / 0° inc
-90° lon / 1° inc
-80° lon / 0° inc
-70° lon / 0° inc
-100° lon / 3° inc
-60° lon / 0° inc
-90° lon / 3° inc
Number of H₀ merge events
102 satellites | 1,296 depot candidates | 7-day mean transfer-cost model